COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	07/29/26	**DATE SUBMITTED:**	07/29/26
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

Tasks & Intended Learning Outcomes (ILOs):
ILO A: Implement a stack using the C++ STL (std::stack).
Importing stack library

```
#include <iostream>
#include <stack>

int main() {

    std::stack<int> stack1, stack2;
```

Checking if the stack is empty

```
//is empty
std::cout<<"Is the stack empty? " << stack1.empty() << std::endl;
```

Pushing a stack a printing the top

```
//push
std::cout<<"========STACK 1========" << std::endl;
stack1.push(10);
std::cout<<"The top stack is " <<  stack1.top() << std::endl;

stack1.push(9);
std::cout<<"The top stack is " <<  stack1.top() << std::endl;

stack1.push(8);
std::cout<<"The top stack is " <<  stack1.top() << std::endl;

stack1.push(7);
std::cout<<"The top stack is " <<  stack1.top() << std::endl;

stack1.push(6);
std::cout<<"The top stack is " <<  stack1.top() << std::endl;
```

Popping a stack

```
//pop
std::cout<<"Testing the pop: " <<  stack1.top() << std::endl;
stack1.pop();
std::cout<<"The top stack is " <<  stack1.top() << std::endl;
std::cout<<"Size of the stack: " << stack1.size() << std::endl;
```

Emplacing a stack

```
//emplace
stack1.emplace(69420);
std::cout<<"The top stack is " << stack1.top() << std::endl;
```

Making another stack to use swap

```
//stack 2 to use swap
std::cout<<"========STACK 2========" << std::endl;
stack2.push(1);
std::cout<<"The top stack is " <<  stack2.top() << std::endl;

stack2.push(2);
std::cout<<"The top stack is " <<  stack2.top() << std::endl;

stack2.push(3);
std::cout<<"The top stack is " <<  stack2.top() << std::endl;

stack2.push(4);
std::cout<<"The top stack is " <<  stack2.top() << std::endl;

stack2.push(5);
std::cout<<"The top stack is " <<  stack2.top() << std::endl;

//swap
stack1.swap(stack2);
```

Displaying all values in the stack

```cpp
//display entire stack 1
std::cout<<"========STACK 1========" << std::endl;
std::cout<<"Displaying Entire Stack1 after swapping:" << std::endl;
while (!stack1.empty()) {
    std::cout << stack1.top() << "\n";
    stack1.pop();
}
```

Output:

```
Testing the stack STL
Is the stack empty? 1
========STACK 1========
The top stack is 10
The top stack is 9
The top stack is 8
The top stack is 7
The top stack is 6
Testing the pop: 6
The top stack is 7
Size of the stack: 4
The top stack is 69420
========STACK 2========
The top stack is 1
The top stack is 2
The top stack is 3
The top stack is 4
The top stack is 5
========STACK 1========
Displaying Entire Stack1 after swapping:
5
4
3
2
1
```

ILO B.1: Implement a stack using a Static Array.
Global declaration

```
//global declarations
#define maxCap 100

int StackArr[maxCap];
int top = -1;    // stack starts empty
int newData;
```

Prototype functions

```
//prototype functions
void push();
void pop();
void displayStack();
void angat();
bool isEmpty();
bool isFull();
```

Check if empty/full

```
//check if is empty
bool isEmpty() {
    if(top == -1) return true;
    return false;
}

bool isFull() {
    if(top == maxCap -1) return true;
    return false;
}
```

Pushing a stack

```
void push() {
    //error checking
    if(isFull()){
        std::cout<<"Stack overflow" << std::endl;
        return;
    }

    //pushing to the stack
    std::cout<<"\nEnter a new value: " << std::endl;
    std::cin>>newData;
    StackArr[++top] = newData;
}
```

Popping a stack

```
void pop(){
    //error checking
    if(isEmpty()){
        std::cout<<"Stack Underflow"<<std::endl;
        return;
    }

    //display the value that we are going to pop
    std::cout<<"Popping: " << StackArr[top]<<std::endl;
    //decrement the top value from the stack
    top--;
}
```

Printing the top element/stack

```
void angat(){
    //error catching
    if(isEmpty()){
        std::cout<<"The stack is empty"<<std::endl;
        return;
    }
    std::cout<<"Top element is: "<<StackArr[top]<<std::endl;
}
```

Displaying the entire stack base LIFO

```
void displayStack(){
    //error catching
    if(isEmpty()){
        std::cout<<"Stack underflow"<<std::endl;
        return;
    }

    //display using normal for loop
    std::cout<<"Stack elements: "<<std::endl;
        for(int i = top; i >= 0; i--){
            std::cout<<StackArr[i]<<std::endl;
        }
}
```

Using those function in the main function

```
int main() {

    //main program
    int choice;

    while(true){
        std::cout<<"<<<======Stack Operations=====>>>" << std::endl;
        std::cout<<"1. Push\n2. Pop\n3. Top\n4. isFull\n5. isEmpty\n6. displayStack"<<std::endl;
        std::cout << "\nEnter your choice: ";
        std::cin>>choice;


        switch(choice){
            case 1: push();
            break;
            case 2: pop();
            break;
            case 3: angat();
            break;
            case 4: std::cout<<"Is stack full? " << isFull() << std::endl;
            break;
            case 5: std::cout<<"Is stack empty? "<< isEmpty() << std::endl;
            break;
            case 6: displayStack();
            break;
            default: std::cout<<"Invalid choice"<<std::endl;
        }
    }
}
```

Output:

```
Enter a new value:
5
<<<======Stack Operations=====>>>
1. Push
2. Pop
3. Top
4. isFull
5. isEmpty
6. displayStack

Enter your choice: 1

Enter a new value:
67
<<<======Stack Operations=====>>>
1. Push
2. Pop
3. Top
4. isFull
5. isEmpty
6. displayStack

Enter your choice: 6
Stack elements:
67
5
4
3
2
1
<<<======Stack Operations=====>>>
1. Push
2. Pop
3. Top
4. isFull
5. isEmpty
6. displayStack
```

Submission Requirements:
Documentation: Please use the official Activity Template for your write-up.

Source Code: Submit your C++ file named as LastName_Seatwork4_1.cpp.

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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